

Fuels and Combustion

7.1 Introduction

In the science of heat power engineering the phenomenon of combustion plays a very important part. The term combustion refers to a rapid chemical reaction, usually accompanied by a flame, which occurs between fuels and oxygen. The oxygen is usually supplied by atmospheric air except in rockets where it is carried in the form of liquid oxygen or some oxygen carrier such as hydrogen peroxide or nitric acid. The molecules of fuel and oxygen have a certain 'bonded' energy between their constituent atoms. This energy, which may be called *chemical energy*, is a form of internal energy. In combustion the interatomic bonds of the molecules of fuel and oxygen are broken and rearrangement of atoms in new molecular combination takes place. During this process energy is released to the surroundings in the form of heat and to that extent the chemical energy of the new molecules is lowered. Combustion is therefore an exothermic reaction. In combustion, though old molecules may disappear and form new molecules by rearrangement of atoms, the total number of atoms of each element remains unchanged, following the law of conservation of mass.

In chemical reaction the mixture of substances before the reaction is called the *reactants* and mixture after the reaction is called the *products*. The branch of chemistry which deals with the numerical proportions in which substances react is called stoichiometry.

In processes involving no chemical reaction the chemical energy is not taken into account as it is same before and after the process.

In heat power engineering we are concerned with combustion of various types of fuels, solid, liquid and gas, in various types of equipment such as boilers, internal combustion engines, gas turbines, rockets, etc. The main problems involved are the quantity of air required for combustion and completeness of combustion. The minimum air required for combustion can be calculated if the composition of fuel is known. The fact whether combustion is complete or incomplete and the actual air-fuel ratio used can be determined from the experimental analysis of the products of combustion. The laws of thermodynamics are applicable to combustion and its problems also. The First

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Law helps in the determination of heat transfer, and the maximum temperature reached in the system, whereas the Second Law helps to predict the possibilities of chemical reactions to occur.

7.4 Composition of air

The atmospheric air contains nitrogen and oxygen and traces of other inert gases. For combustion calculations inert gases are considered as nitrogen. The following composition of air may be taken as sufficiently accurate.

	By mass (%)	By volume (%)
Oxygen	23.3	21
Nitrogen	76.7	79

Further approximation is done by taking mass analysis of air as oxygen 23 per cent and nitrogen 77 per cent.

Thus 1 kg of O_2 + 3.32 kg of N_2 = 4.32 kg of air
and 1 mol of O_2 + 3.76 mol of N_2 = 4.76 mol of air

The molecular mass of air is taken as 29.

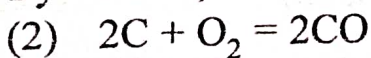
7.5 Combustion equations

The following chemical equations apply to combustion problems in boilers and engines :



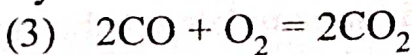
By mass, 12 kg + 32 kg = 44 kg or $1 + \frac{8}{3} = \frac{11}{3}$

By volume, 1 mol + 1 mol = 2 mol



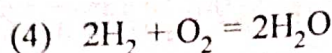
By mass, 24 kg + 32 kg = 56 kg or $1 + \frac{4}{3} = \frac{7}{3}$ (7.2)

By volume, 2 mol + 1 mol = 3 mol



By mass, 56 kg + 32 kg = 88 kg or $1 + \frac{4}{7} = \frac{11}{7}$ (7.3)

By volume, 2 mol + 1 mol = 3 mol

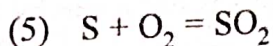


By mass, 4 kg + 32 kg = 36 kg

$$\text{or } 1 + 8 = 9$$

By volume, 2 mol + 1 mol = 2 mol

(7.4)

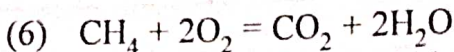


By mass, 32 kg + 32 kg = 64 kg

$$\text{or } 1 + 1 = 2$$

By volume 1 mol + 1 mol = 1 mol

(7.5)



By mass, 16 kg + 64 kg = 44 kg + 36 kg

$$\text{or } 1 + 4 = 2.75 + 2.25$$

By volume, 1 mol + 2 mol = 1 mol + 2 mol

(7.6)

Note.— Contraction in volume takes place in the equations (1), (2), (3), (4) and (5).

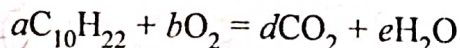
The results of the above equations are summarized in the Tables 7.6.

Table 7.6 Mass of oxygen required and products of combustion

Constituents (a)	Oxygen required kg/kg of constituent (b)	Products of combustion, kg/kg of constituent (c) = (a) + (b)		
		CO ₂	H ₂ O	SO ₂
C	8/3	11/3	—	—
CO	4/7	11/7	—	—
H ₂	8	—	9	—
S	1	—	—	2
CH ₄	4	11/4	9/4	—

(7) Combustion equation for any hydrocarbon

Let a combustion equation for any hydrocarbon C₁₀H₂₂ be required with complete combustion.



Equating C, H and O on the two sides of the equation

$$a \times 10 = d \quad \therefore d = 10a$$

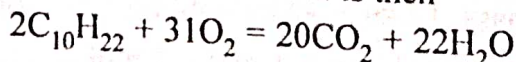
$$a \times 22 = e \times 2 \quad \therefore e = 11a$$

$$b \times 2 = d \times 2 + e \times 1 = 10a \times 2 + 11a \times 1 = 31a, \quad \therefore b = \frac{31}{2}a$$

Taking $a = 1$, we get, $e = 11$, $d = 10$ and $b = \frac{31}{2}$

To avoid fraction take $a = 2$ then, $e = 22$, $d = 20$ and $b = 31$

$\therefore$ The equation of combustion is then



(7.7)

7.6 Excess air and air fuel ratio

The minimum air required for complete combustion is known as the *theoretical or stoichiometric air*. Any air in excess of this is called the *excess air*. In practical combustion systems if only minimum air is supplied the combustion would be

7.7 Minimum air required for complete combustion

incomplete because due to limited reaction time available there is imperfect mixing of fuel and air. Also the inert gas molecules obstruct the reaction between the active molecules of fuel and oxygen. The quantity of excess air depends on the type of fuel and method of combustion used and may be from 25 to 100 per cent. Excess air ensures better combustion but at the same time cools the furnace and increases heat loss in the exhaust gases.

The air-fuel ratio (AF) is defined as the ratio of the mass of the air to fuel. If theoretical (or chemically correct) air is supplied we have stoichiometric air-fuel ratio. The inverse of air fuel ratio is fuel-air ratio (FA).

The percentage of excess air is given by the following formula.

$$\text{Percentage excess air} = \frac{\text{actual A/F ratio} - \text{stoichiometric A/F ratio}}{\text{stoichiometric A/F ratio}} \quad (7.8)$$

7.7 Minimum air required for complete combustion

From the previous article it is seen that an adequate supply of oxygen is essential for complete combustion and this is obtained from atmospheric air along with nitrogen. For all practical purposes we can consider air to contain oxygen by mass 23 per cent and by volume 21 per cent and nitrogen by mass 77 per cent and by volume 79 per cent. For calculating the amount of oxygen required, the oxygen content of the fuel itself must be considered as it also takes part in combustion. Thus net oxygen required will be less. The calculations of air required are given in the Table 7.7 for a fuel having the following analysis by mass, C = 88.4%, H₂ = 3.6%, O₂ = 2.6% N₂ = 0.6%, S = 0.8% H₂O = 0.8% and ash 3.2%.

Table 7.7. Mass of air required for a fuel

Mass of constituents per kg of fuel (a)	Oxygen required per kg of constituents (b)	Oxygen required per kg of fuel (c) = (a) × (b)
C = 0.884	8/3	2.357
H ₂ = 0.036	8	0.288
O ₂ = 0.026	—	-0.026
N ₂ = 0.006	—	—
S = 0.008	1	0.008
H ₂ O = 0.008	—	—
ash = 0.032	—	—
Total 1.000	—	Σ(c) = 2.627

$$\therefore \text{Mass of air required (theoretical)} = \frac{2.627 \times 100}{23} = 11.42 \text{ kg}$$

7.8 Wet and dry analysis of products of combustion

The analysis of the products of combustion can be given either on mass basis or volume basis. The products of combustion in hot condition have water in the vaporized form. The analysis made taking water vapour into account is

called the wet analysis. If the analysis is made on the assumption that vapour has been condensed and removed it is called dry analysis. It must be emphasised here that merely cooling the products does not separate water vapour entirely. There will be some vapour still left at partial pressure corresponding to the temperature of products after cooling. The water vapour can however be removed entirely by using drying agent.

7.9 Analysis of flue gases

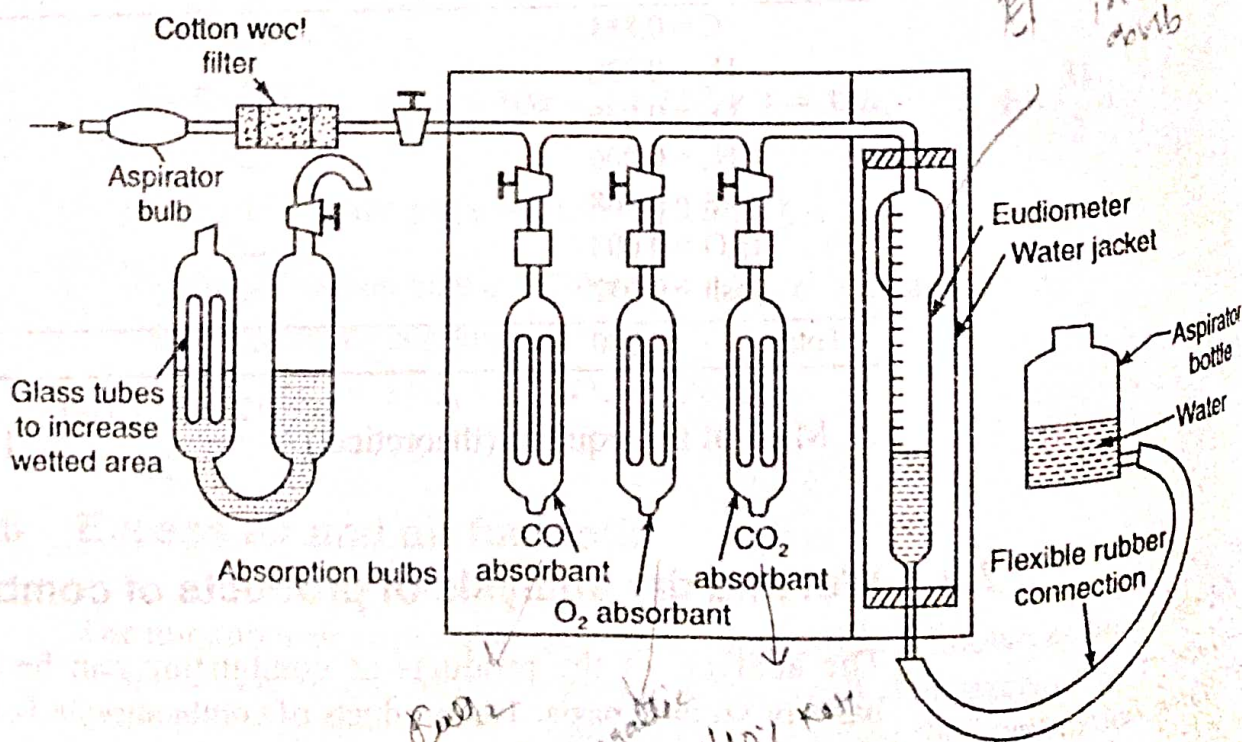
The volumetric analysis of dry products of combustion from a boiler or an engine is done by means of Orsat apparatus and by infra red gas analyser.

7.9.1 Orsat apparatus

The most common method of analysing flue gases is by Orsat apparatus (Fig. 7.1). It consists of a graduated eudiometer tube connected to an aspirator-bottle containing water and three double absorption flasks of CO_2 , O_2 and CO respectively. The front flasks are packed with glass tubes to increase the wetted surface.

The first flask next to eudiometer tube contains 40 per cent of KOH solution which would absorb twenty times its own volume of CO_2 . The second flask contains pyrogalllic acid and KOH . This mixture may be made by mixing 5 gm of powdered pyrogalllic acid with 100 cc of 40 per cent KOH solution which would absorb twice its own volume of O_2 . The third flask contains a solution of 5 gm of copper oxide in 100 cc of commercial HCl , which would absorb CO equal to its own volume.

7.1 Orsat apparatus



7.9 Analysis of fuel gases

Gas to be analysed may be collected directly into eudiometer tube after removing air, or it may be collected in jars on salt water (pure water may absorb some gases). After 100 cc of gas is taken into eudiometer tube and it is forced into the first flask by raising the bottle where CO_2 is absorbed. The liquid in the flask is then levelled back by slowly lowering the aspirator bottle and the reading of eudiometer tube is taken. The difference in readings gives percentage of CO_2 by volume. The process is repeated with second and third flasks to determine O_2 and CO percentages.

Additional flasks may be provided for the determination of hydrocarbons and hydrogen. The balance is assumed N_2 .

The flue gases will have H_2O vapour due to presence of hydrogen in fuel. But in Orsat apparatus only dry flue gas will be collected as H_2O vapour will condense at room temperature. Therefore the percentage CO_2 obtained by Orsat analysis will slightly more than actual.

7.11 Conversion of volumetric analysis to analysis by mass

Let the volumetric analysis of dry flue gases, as measured by Orsat apparatus, burning the fuel mentioned in article 7.7, be $\text{CO}_2 = 7.79\%$, $\text{CO} = 0.535\%$, $\text{O}_2 = 11.6\%$ and $\text{N}_2 = 80.075\%$. The volumetric analysis can be converted to mass analysis as in Table 7.8.

The principle behind conversion of mass analysis into volume analysis and vice-versa mercury should be clearly understood. 22.41 m^3 (one mole) of all gases at N.T.P. (760 mm of mercury and 0°C) weigh equal to their emolecular weight. Therefore, dividing by molecular weight we get mole or proportional volume. Similarly, by multiplying mole (or volume) by molecular weight we get proportional mass.

Table 7.8. Conversion of volumetric analysis into mass analysis

Volume of constituents per mole of dfg (a)	Mol wt. (b)	Proportional mass of constituents (c) = (a) $\times$ (b)	Mass per kg of flue gases (d) = (c)/ Σ (c)	Mass of carbon in kg per kg of flue gases (e)
$\text{N}_2 = 80.075$	28	$28 \times 80.075 = 2242$	0.75461, 75.46%	
$\text{CO} = 0.535$	28	$28 \times 0.535 = 15$	0.00505, 0.505%	$\frac{0.00505 \times 12}{28} = 0.002164$
$\text{CO}_2 = 7.790$	44	$44 \times 7.790 = 343$	0.11551, 11.55%	$\frac{0.11551 \times 12}{44} = 0.03150$
$\text{O}_2 = 11.600$	32	$32 \times 11.600 = 371$	0.12491, 12.49%	
Total = 100.000		Σ (c) = 2971	Σ (d) 1.00008, 100.005	0.033664

7.12 Mass of dry flue gases per kg of fuel burnt

In flues gases only CO_2 and CO contain carbon. From equations (7.1) and (7.2) we know that 44 kg of CO_2 contains 12 kg of carbon, and 28 kg of CO contains 12 kg of carbon. From this, carbon content per kg of flues gases is found as in Table 7.7. Knowing total carbon per kg of fuel, i.e. carbon content of fuel, the mass of dry flue gases can be found as follows :

Mass of dry gases per kg of fuel

mbustion

$$\frac{\text{mass of carbon per kg of fuel}}{\text{mass of carbon per kg of gases}} = \frac{0.884}{0.033664} = 26.26 \text{ kg}$$

(7.9)

7.13 Mass of excess air supplied

Minimum O_2 required is on the basis of complete combustion, i.e. whole C should be burnt to CO_2 and no CO should be present in the flue gases. If in a case, CO and O_2 are present in the gas analysis, whole of this O_2 is not "excess". Excess O_2 is found by subtracting the amount of O_2 required for burning CO to CO_2 .

Again taking previous example dealt in articles 7.7 and 7.11.

Mass of flue gases per kg of fuel = 26.26 kg

From table 7.7, Mass of CO per kg of fuel = $26.26 \times 0.00505 = 0.1326 \text{ kg}$

Mass of unused O_2 per kg of fuel = $26.26 \times 0.1249 = 3.27987 \text{ kg}$

Mass of O_2 required for burning of CO

$$= 0.1326 \times \frac{4}{7} = 0.07577 \text{ kg}$$

Excess O_2 per kg of fuel = $3.27987 - 0.07577 = 3.2041 \text{ kg}$

$$\therefore \text{Excess air} = \frac{3.2041 \times 100}{23} = 13.93 \text{ kg/kg of fuel}$$

$$\text{or Percentage excess air} = \frac{13.93}{11.42} = 122 \text{ per cent.}$$